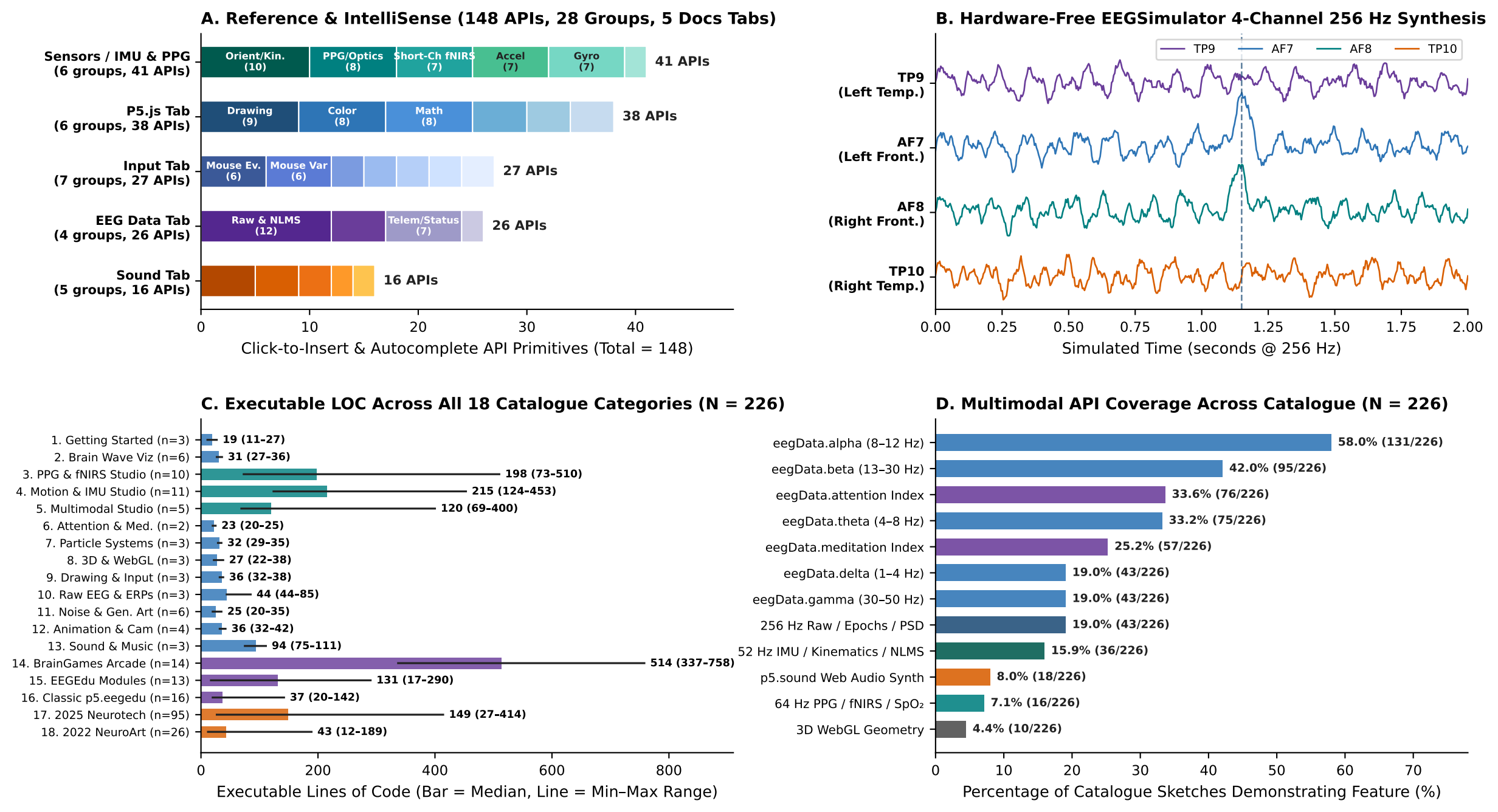

A. Reference & IntelliSense (148 APIs, 28 Groups, 5 Docs Tabs)
Sensors / IMU & PPG
(6 groups, 41 APIs)
Orient/Kin.
(10)
PPG/Optics
(8)
Short-Ch fNIRS
(7)
Accel
(7)
Gyro
(7)
41 APIs
P5.js Tab
(6 groups, 38 APIs)
Drawing
(9)
Color
(8)
Math
(8)
38 APIs
Input Tab
(7 groups, 27 APIs)
Mouse Ev.
(6)
Mouse Var
(6)
27 APIs
EEG Data Tab
(4 groups, 26 APIs)
Raw & NLMS
(12)
Telem/Status
(7)
26 APIs
Sound Tab
(5 groups, 16 APIs)
16 APIs
0
10
20
30
40
Click-to-Insert & Autocomplete API Primitives (Total = 148)
B. Hardware-Free EEGSimulator 4-Channel 256 Hz Synthesis
TP9
AF7
AF8
TP10
TP9
(Left Temp.)
AF7
(Left Front.)
AF8
(Right Front.)
TP10
(Right Temp.)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Simulated Time (seconds @ 256 Hz)
C. Executable LOC Across All 18 Catalogue Categories (N = 226)
1. Getting Started (n=3)
19 (11–27)
2. Brain Wave Viz (n=6)
31 (27–36)
3. PPG & fNIRS Studio (n=10)
198 (73–510)
4. Motion & IMU Studio (n=11)
215 (124–453)
5. Multimodal Studio (n=5)
120 (69–400)
6. Attention & Med. (n=2)
23 (20–25)
7. Particle Systems (n=3)
32 (29–35)
8. 3D & WebGL (n=3)
27 (22–38)
9. Drawing & Input (n=3)
36 (32–38)
10. Raw EEG & ERPs (n=3)
44 (44–85)
11. Noise & Gen. Art (n=6)
25 (20–35)
12. Animation & Cam (n=4)
36 (32–42)
13. Sound & Music (n=3)
94 (75–111)
14. BrainGames Arcade (n=14)
514 (337–758)
15. EEGEdu Modules (n=13)
131 (17–290)
16. Classic p5.eegedu (n=16)
37 (20–142)
17. 2025 Neurotech (n=95)
149 (27–414)
18. 2022 NeuroArt (n=26)
43 (12–189)
0
200
400
600
800
Executable Lines of Code (Bar = Median, Line = Min–Max Range)
D. Multimodal API Coverage Across Catalogue (N = 226)
eegData.alpha (8–12 Hz)
58.0% (131/226)
eegData.beta (13–30 Hz)
42.0% (95/226)
eegData.attention Index
33.6% (76/226)
eegData.theta (4–8 Hz)
33.2% (75/226)
eegData.meditation Index
25.2% (57/226)
eegData.delta (1–4 Hz)
19.0% (43/226)
eegData.gamma (30–50 Hz)
19.0% (43/226)
256 Hz Raw / Epochs / PSD
19.0% (43/226)
52 Hz IMU / Kinematics / NLMS
15.9% (36/226)
p5.sound Web Audio Synth
8.0% (18/226)
64 Hz PPG / fNIRS / SpO₂
7.1% (16/226)
3D WebGL Geometry
4.4% (10/226)
0
10
20
30
40
50
60
70
Percentage of Catalogue Sketches Demonstrating Feature (%)